SentinelIQ 01 — Vision & Problem Statement

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SentinelIQ is an AI-assisted SOC copilot that ranks security alerts, explains the risky ones with MITRE ATT&CK references, and leaves every decision to a human analyst.

Problem

Security teams receive more alerts than analysts can investigate, so real incidents wait in the same queue as false positives. Most tools give a score without a reason, which forces the analyst to re-investigate every alert from scratch.

- **Volume:** alert queues grow faster than headcount, and analysts triage under time pressure.
- **Noise:** a large share of alerts are benign, so genuine attacks are easy to miss.
- **Opacity:** a bare risk score does not say which attacker behaviour it resembles or what to check next.

Vision

SentinelIQ ingests security events, scores them for anomalies, and writes a short explanation for each flagged event that cites the relevant MITRE ATT&CK technique. Analysts review the queue and make the final call.

The rule it never breaks: SentinelIQ recommends and explains; it never blocks, deletes or remediates anything on its own. Every recommendation is stored with its evidence, and every human decision is logged.

Target users

Role	Goal	Main pain point
SOC analyst (primary)	Decide quickly which alerts are real and what to do next	Too many alerts and too little context per alert
SOC lead	See trends and review analyst decisions for quality	No audit view of why alerts were closed

Role	Goal	Main pain point
Administrator	Manage accounts and roles safely	Unclear who can see or change what

Success metrics

Targets are proposed and will be revisited after the Phase 4 baseline is measured.

Metric	Proposed target	How it is measured
Attack recall	90% or higher on the held-out test set	Confusion matrix per attack class
False-positive rate	5% or lower at the chosen threshold	Same held-out set
Explanation accuracy	80% or higher of cited techniques match ground truth	Small hand-labeled evaluation set
Injection resistance	Report the share of malicious log lines that fail to change output	Fixed adversarial test set
Ingestion latency	95% of requests under 300 ms	Load test in Phase 7
Setup time	A new machine runs the full demo in under 10 minutes	Timed clean-checkout run

Scope boundaries

In scope for v1

- Ingestion of two event sources into one schema
- ML anomaly scoring with versioned models
- Cited explanations grounded in MITRE ATT&CK
- Analyst dashboard, decisions and audit log
- Role-based access, tests, CI and containerized deployment

Out of scope for v1

- Live connectors to real company systems
- Automated response or blocking

- Multi-tenant hosting, billing and single sign-on

The public dataset contains network flows, not raw application logs, so the authentication-log source is synthetic. Documents 04 and 06 explain how to present this honestly.

Assumptions and risks

Risk	Impact	Mitigation
Public flow data does not look like real production traffic	Metrics may overstate real-world performance	State the limit openly; evaluate on held-out classes; avoid claims about live traffic
LLM produces a confident but wrong explanation	Analyst is misled	Cite only retrieved techniques; show confidence and sources; keep the human decision
Scope grows beyond what one person can finish	No complete demo	Fixed v1 scope in Document 02; should-have items only after must-haves pass
Hosted LLM cost or outage	Explanations unavailable	Graceful degradation; cache explanations; keep a local-model fallback option
Log content attacks the LLM layer	Manipulated output	Controls in Document 03; adversarial tests reported in the README